EXPERIMENT 4:

Setup and Configuration of a LAN (Local Area Network) Using a Switch and Ethernet Cables

AIM :

To setup and configure a Local Area Network (LAN) using a switch and Ethernet cables.

PROCEDURE

1. **Plan and design** an appropriate network topology based on the network requirements and equipment location.
2. Take **4 computers, 1 network switch** with 8, 16, or 24 ports, and **4 Ethernet cables**.
3. Connect all four computers to the network switch using Ethernet cables. 4. **Assign IP addresses** to all PCs:
    - Log in to the client computer.
    - Go to **Network and Internet Connections**.
    - Right-click **Ethernet/Local Area Connection** and select **Properties**.
    - Select **Internet Protocol (TCP/IPv4)** and click **Properties**.
    - Select **Use the following IP address** and enter the required IP address.
5. Assign the following IP addresses:

Device	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0

6. **Configure the network switch**:
    - Connect the computer to the switch using an Ethernet cable.
    - Open a web browser and enter the switch IP address.
    - Log in using the username and password.
    - Configure the basic switch settings.
    - Assign the switch IP address as **10.1.1.5** with subnet mask **255.0.0.0**. 7. Check the connectivity between the switch and other computers using the **ping command** in the Command Prompt.
8. Select a folder, open **Properties**, go to the **Sharing** tab, and share the folder with everyone on the same LAN.
9. Access the shared folder from the other computers connected to the network.

RESULT

Thus, a LAN was successfully setup and configured using a switch and Ethernet cables, and connectivity between the connected computers was verified using the ping command.